CHAPTER 9

INTERNET

CHAPTER ESSENCE AT A GLANCE:
The Internet integrates distributed heterogeneous networks through standardized transmission protocols, domain routing, and web services to enable universal communications.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **ARPANET Lineage and Protocol Federation:** Originating from DARPA defense research in 1969, the Internet unified disparate networks using the standardized Transmission Control Protocol and Internet Protocol (TCP/IP) suite.

• **IP Address Geometries and Namespace Resolution:** Internet protocol numbering transitions from 32-bit IPv4 addresses to massive 128-bit IPv6 spaces, mapped to human-readable domain names via the hierarchical Domain Name System (DNS).

• **Application Protocols and Web Architecture:** High-level protocols deliver services; HTTP and secure HTTPS transfer hypertext pages, FTP manages bulk file transfers, and SMTP handles email transmission.

• **Cyber Threat Vectors and Defense Safeguards:** Digital assets require defense against malicious viruses, self-replicating worms, trojan backdoors, and fraudulent phishing through hardware firewalls, antivirus suites, and encrypted channels.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Address Space} = \frac{\text{IPv4: 32 Bits (4 Octets)} \rightarrow 2^{32} \approx 4.3 \times 10^{9} \text{ Hosts}}{\text{IPv6: 128 Bits (8 Hex Groups)} \rightarrow 2^{128} \approx 3.4 \times 10^{38} \text{ Hosts}}$$

IPv4 versus IPv6 Address Space Dimensions

$$\text{URL Structure} = \text{Protocol } (\textbf{https://}) + \text{Domain } (\textbf{site.org}) + \text{Port } (:443) + \text{Path } (/\textbf{doc.html})$$

Uniform Resource Locator Component Structure

$$\text{Core Protocols} = \text{TCP (Reliable Byte Stream)} + \text{IP (Packet Routing)} + \text{DNS (Name Translation)}$$

Fundamental Internet Protocol Suite Triad

$$\text{Defense Matrix} = \frac{\text{Threats: Viruses, Worms, Trojans, Phishing, Denial of Service}}{\text{Defenses: Packet Firewalls, Antivirus, SSL / TLS, Strong Passwords}}$$

Cybersecurity Threat Vectors vs Protective Shields

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** A computer virus requires an executable host file to propagate, whereas a computer worm is an autonomous self-replicating program that spreads across networks.

• **Alert:** Phishing attacks rely on fraudulent emails and deceptive lookalike websites to trick users into divulging confidential login passwords and financial credentials.

• **Alert:** IPv4 addresses are 32 bits divided into 4 decimal octets, while IPv6 addresses are 128 bits divided into 8 hexadecimal groups separated by colons.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is the Internet? Name the historical network project that laid the foundation for the modern Internet. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Internet Definition: A global network of interconnected computer networks utilizing standardized TCP/IP protocols to communicate and exchange data worldwide. **[1 Mark]**

• Historical Foundation: ARPANET (Advanced Research Projects Agency Network) initiated by the United States Department of Defense in 1969. **[1 Mark]**

2. Distinguish between IPv4 address and IPv6 address with examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• IPv4 Address: 32-bit numerical identifier represented as 4 decimal octets separated by dots (e.g. 192.168.1.1), providing approximately 4.3 billion addresses. **[1 Mark]**

• IPv6 Address: 128-bit numerical identifier represented as 8 groups of hexadecimal digits separated by colons (e.g. 2001:0db8:85a3::8a2e), solving address exhaustion. **[1 Mark]**

3. Explain the essential components of a Uniform Resource Locator (URL) with an illustrative example. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Example URL: https://www.dhsekerala.gov.in/exams/results.html. **[0.5 Mark]**

• Protocol: https:// specifies the communication rules (Hypertext Transfer Protocol Secure). **[0.75 Mark]**

• Domain Name: www.dhsekerala.gov.in represents the unique host server name on the Internet. **[0.75 Mark]**

• Directory Path and File Name: /exams/ specifies the directory folder hierarchy on the server, and results.html is the specific target web document. **[1 Mark]**

4. Differentiate between a computer virus, a computer worm, and a Trojan horse. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Computer Virus: A malicious program that attaches itself to legitimate executable host programs and reproduces when the infected program is executed, damaging files. **[1 Mark]**

• Computer Worm: A standalone self-replicating malware program that actively consumes network bandwidth and memory without requiring an existing host file. **[1 Mark]**

• Trojan Horse: A deceptive program that disguises itself as useful legitimate software while secretly creating backdoors for unauthorized remote access. **[1 Mark]**

5. Explain how the Domain Name System (DNS) translates a website domain name into a numerical IP address. [Score: 3 Marks]

Answer & Key Valuation Steps:

• DNS Role: Operates as the distributed directory of the Internet, translating human-friendly names (like www.google.com) into machine-routable IP addresses. **[1 Mark]**

• Resolution Hierarchy: The browser queries the local cache, then the ISP recursive resolver, which queries the Root DNS server ('.'), the Top-Level Domain (TLD) server ('.com'), and finally the Authoritative Name Server. **[1 Mark]**

• Response Delivery: The authoritative server returns the exact IP address to the resolver, which caches it and delivers it to the browser to initiate the HTTP connection. **[1 Mark]**

6. What is phishing? State three essential safety guidelines to protect against online financial fraud. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Definition: A deceptive social engineering cyber crime where attackers pose as legitimate financial institutions via spoofed emails and fake websites to steal passwords and credit card details. **[1.5 Marks]**

• Three Safety Guidelines: Never click on unverified links received via unsolicited emails or messages; always verify the HTTPS lock icon in the browser address bar; never disclose OTPs, PINs, or CVV numbers. **[1.5 Marks]**

7. Explain the role of Firewalls and Antivirus software in establishing multi-layered computer security. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Firewall Architecture: A network security system (hardware or software) that monitors and filters incoming and outgoing network traffic based on predefined security rules. **[1 Mark]**

• Firewall Defensive Action: Blocks unauthorized port scans, prevents malicious hackers from establishing illicit connections, and shields internal private networks from external threats. **[1 Mark]**

• Antivirus Software: Utility software that scans local storage drives and incoming email attachments to detect, quarantine, and eliminate virus signatures and spyware. **[1 Mark]**

• Synergistic Protection: The firewall protects the network perimeter from external intrusion, while antivirus software neutralizes infections that penetrate through physical media or downloads. **[1 Mark]**